

CSD, Rennes

TD 2

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1 Problem 1 : Your first state-feedback control

Linearization

1.1 Set $u_e = 2$ and study the equilibrium

We have

$$\begin{cases} -x_1 + x_1 x_2 = 0 \\ -x_2 + x_1 x_2 = u_e \end{cases} \Leftrightarrow \begin{cases} x_1 (x_2 - 1) = 0 \\ x_2 (x_1 - 1) = u_e. \end{cases}$$

It yields two possible equilibrium points

$$x_e^1 = \begin{bmatrix} 0 \\ -u_e \end{bmatrix} = \begin{bmatrix} 0 \\ -2 \end{bmatrix}, \quad x_e^2 = \begin{bmatrix} u_e + 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 3 \\ 1 \end{bmatrix}.$$

1.2 Determine the linearized models

We have

$$\dot{\tilde{x}} = \underbrace{f(x_e, u_e)}_0 + \underbrace{\frac{\partial f(x_e, u_e)}{\partial x}}_A \tilde{x} + \underbrace{\frac{\partial f(x_e, u_e)}{\partial u}}_B \tilde{u} = A(x_e, u_e) \tilde{x} + B(x_e, u_e) \tilde{u},$$

where

$$A(x_e, u_e) = \begin{bmatrix} -1 + x_{e,2} & x_{e,1} \\ -x_{e,2} & 1 - x_{e,1} \end{bmatrix}, \quad B(x_e, u_e) = \begin{bmatrix} 0 \\ 1 \end{bmatrix},$$

and for x_e^1, x_e^2 we compute

$$A^1 = \begin{bmatrix} -3 & 0 \\ 2 & 1 \end{bmatrix}, \quad A^2 = \begin{bmatrix} 0 & 3 \\ -1 & -2 \end{bmatrix}.$$

1.3 Study the stability and controllability

Obviously, the equilibrium x_e^1 is not controllable - x_1 is autonomous. To check, compute

$$\mathcal{C}_1 = \begin{bmatrix} 0 & 0 \\ 1 & 1 \end{bmatrix}.$$

It is also easy to see that it is unstable as, e.g. for $\tilde{x}_1(0) = 0$ we have $\dot{\tilde{x}}_2 = \tilde{x}_2$. To check, the eigenvalues are -3 and 1 .

For the second equilibrium, we have a controllable linearization:

$$\mathcal{C}_2 = \begin{bmatrix} 0 & 0 \\ 3 & -2 \end{bmatrix}.$$

Moreover, it is stable and the eigenvalues of A^2 are $-1 \pm \sqrt{2}i$.

Next, we consider only the second (controllable) linearization and set

$$A = A^2, \quad x_e = x_e^2.$$

Pole placement

1.4 Propose eigenvalues

It is a second-order system. If we set one pole 10 time bigger then another, we can approximate it as a first-order system. Then the transient time will be $-3/\lambda_1$, where λ_1 is the slowest pole. Given the desired time 0.5 months, we compute $\lambda_1 \leq -6$ and $\lambda_2 = -10\lambda_1$.

1.5 Compute K

$K = \text{place}(A, B, [-6 \ -60])$ yields $K = [119 \ 64]$. At the same time,

$$\det \left(sI - \begin{bmatrix} 0 & 3 \\ -1 - k_1 & -2 - k_2 \end{bmatrix} \right) = s^2 + (k_2 + 2)s + 3k_1 + 3$$

and given $(s + 6)(s + 60) = s^2 + 66s + 360$ we get the same.

1.6 Simulink

Simulations

LQ

1.7 Conditions

The first condition is to have (A, B) stabilizable - checked before. The second condition is to find C such that $Q = C^\top C$ and check if (A, C) is observable.

Given $Q = \begin{bmatrix} q_1 & 0 \\ 0 & q_2 \end{bmatrix}$, we have $C = \begin{bmatrix} \sqrt{q_1} & 0 \\ 0 & \sqrt{q_2} \end{bmatrix}$. We have

$$CA = \begin{bmatrix} 0 & 3\sqrt{q_1} \\ -\sqrt{q_2} & -2\sqrt{q_2} \end{bmatrix}$$

and the system is observable if any of q_1 or q_2 is positive.

1.8 Simulink

`K_LQ=lqr(A,B,[q-1, 0; 0, q-2],1)`

1.9 Summary

The bigger is Q the more aggressive are the transients.

Nonlinear

1.10 What is the expression of the control of the nonlinear system ?

We add the linearized control $\tilde{u}$ to the one, around which we linearized, that is u_e , and recall that $\tilde{x}$ is the deviation around the equilibrium x_e

$$u(t) = u_e + \tilde{u}(t) = u_e - K\tilde{x} = u_e - K(x - x_e).$$

1.11 Simulink

Simulations

1.12 Discussion

Linearization allows to stabilize the nonlinear system.

2 Problem 2 : From transfer function to state-feedback control

Rewrite it as

$$\frac{Y(s)}{U(s)} = \frac{s+1}{s^3 + 20s^2 + 100s}.$$

Then the canonical controllable form is

$$A = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & -100 & -20 \end{bmatrix}, \quad B = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}, \quad C = [1 \quad 1 \quad 0].$$

The system is of order 3, and we have to choose three poles.

2.1 First case

Let us approximate it as a second-order system and just send the 3rd pole far to the left. Let the second-order system's poles satisfy $s^2 + 2\xi\omega s + \omega^2 = 0$. For info, the roots are

$$-\omega \left(\xi \pm i\sqrt{1-\xi^2} \right).$$

To have the 5% overshoot, we set $\xi = 0.7$. Then $t_r \approx 3\omega$ and from $t_r = 2$ we obtain $\omega = 1.5$. The last root we choose as -10ω . To get the gains, we compute `K=place(A,B,poles)`.

To get ℓ , we have

$$y_\infty = -C(A - BK)^{-1} B \ell y^r \implies \ell = \frac{-1}{C(A - BK)^{-1} B}.$$

2.2 Second case

We use the same first and second poles, but the 3rd one we choose to compensate the zero, i.e. it equals to -1 . All the rest is the same.